

Falling film evaporation of ammonia on smooth and Low-Fin tube arrays

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ABSTRACT

The heat transfer performance of a falling film evaporator using ammonia was thoroughly investigated by directly comparing it with submerged natural convection/pool boiling, focusing on the effect of tube position in a vertical tube arrangement. Falling film evaporation of ammonia in a vertical tube array was examined at saturation temperatures ranging from 10 to 30 °C, comparing it with submerged natural convection/pool boiling under low heat flux conditions (0.5–18.2 kW/m²). Five stainless steel test tubes having an outer diameter of 15.87 mm were arranged vertically, with one submerged in liquid ammonia and the others exposed to falling film evaporation. Experimental results show that heat transfer coefficients (HTCs) in the falling film evaporation are up to 4.4 times higher than those in the submerged natural convection/pool boiling. The uppermost tube exhibited the lowest HTC among the film-evaporated tubes due to the formation of locally thick liquid films. Empirical correlations were developed to predict the HTC as a function of Reynolds number, Prandtl number, reduced pressure, and dimensionless heat flux, with a mean absolute percentage error of less than 11.1 %. The proposed correlations are applicable to the tested range of 10–30 °C and low heat flux (0.5–18.2 kW/m²). Enhanced low-fin tubes exhibited 10–30 % improved thermal performance over smooth tubes, particularly at higher saturation temperatures, though the performance gain was limited at low temperatures due to hindered liquid spreading by the fin structures. The findings highlight the critical role of tube arrangement, film flow characteristics, and enhanced surfaces in optimizing ammonia falling film evaporators for efficient and environmentally friendly heat pump applications.

1. Introduction

The importance of heat pumps is increasing as a means to reduce greenhouse gas emissions in heating and cooling systems in industrial processes and buildings. Vapor compression heat pumps are replacing traditional fossil fuel boilers, thereby reducing greenhouse gas emissions [1]. Moreover, absorption and adsorption heat pumps utilize waste heat energy, providing environmentally friendly heating and cooling solutions. However, regulations on refrigerants in heat pumps have been strengthened through the Kigali Amendment, establishing a roadmap for phasing out conventional HFC (Hydrofluorocarbon) refrigerants. As synthetic refrigerants face stricter regulations, there is a growing effort to utilize natural refrigerants, such as water, hydrocarbons, and ammonia, in heat pumps. Although ammonia has long been used in fertilizers and the chemical industry, and it has also been partially utilized as a working fluid in industrial heat pumps, its high toxicity and flammability have limited its application in heating and cooling. Nevertheless, with the increasing interest in eco-friendly refrigerants,

ammonia is once again gaining attention for heating and cooling applications in industries and buildings [2].

As the utilization of heat pumps increases, improving their efficiency has become increasingly important [3]. Evaporators are not only a significant component of refrigeration and air conditioning systems, including heat pumps, but are also importantly used in various energy-related industries such as desalination and chemical processes [4]. Falling film evaporators are being adopted more widely due to their higher heat transfer efficiency compared to flooded evaporators. These evaporators achieve heat transfer through conduction via a thin liquid film and evaporation at the liquid–gas interface, resulting in a high heat transfer coefficient (HTC). Additionally, since they do not require complete immersion of the tube, they can reduce the amount of refrigerant used [5]. Although falling film evaporators exhibit excellent heat transfer performance, the refrigerant must be continuously supplied to wet the evaporator tubes, and their performance is influenced by numerous factors, such as the refrigerant flow rate and the spacing between tubes. Extensive research has been conducted to examine the effects of these variables, specifically investigating parameters such as

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Nomenclature			
A	area (m^2)	q''	heat flux (W/m^2)
A_i	inner surface area (m^2)	R_a	arithmetic average roughness (m)
A_o	outer surface area (m^2)	Re_D	Reynolds number
A_t	total surface area (m^2)	Re_f	film Reynolds number
c_p	isobaric specific heat (J/kgK)	s	spacing between injection holes
D_i	tube inner diameter (m)	t_w	tube thickness (m)
D_o	tube outer diameter (m)	T	temperature (K)
D_f	outmost diameter including fins (m)	T_i	inlet temperature (K)
f	friction factor	T_o	outlet temperature (K)
g	gravitational acceleration (m/s^2)	t_w	tube wall thickness (m)
HTC	heat transfer coefficient	U	overall heat transfer coefficient ($\text{W}/\text{m}^2\text{K}$)
h_i	convective heat transfer coefficient in pipe ($\text{W}/\text{m}^2\text{K}$)	<i>Greek letters</i>	
h_o	evaporation heat transfer coefficient ($\text{W}/\text{m}^2\text{K}$)	Γ	liquid mass flow rate per unit length of tube (each side) (kg/ms)
h_f	fin height (m)	ΔT_{lm}	log mean temperature difference (K)
h_{fg}	heat of vaporization (J/kg)	θ	dimensionless heat flux
k	thermal conductivity (W/mK)	λ_d	most dangerous wavelength (m)
L	tube length (m)	μ	viscosity ($\text{Pa}\cdot\text{s}$)
MAPE	mean absolute percentage error	ρ	density (kg/m^3)
$\dot{m}$	mass flow rate (kg/s)	σ	surface energy (N/m)
Nu_D	Nusselt number	<i>Subscripts</i>	
P	pressure (Pa)	D	diameter
P_{crit}	critical pressure (Pa)	l	liquid
P_{red}	reduced pressure (Pa), $= P/P_{crit}$	sat	saturation
Pr	Prandtl number	v	vapor
Q	heat transfer rate (W)		

refrigerant flow rate [6–8], heat flux [9,10], saturation temperature [11], and nozzle injection method [12]. However, the effects of individual parameters are inconsistent owing to not only the complicated heat transfer mechanism of falling film evaporation but also the interconnection between these parameters. For instance, regarding the effect of the flow rate, Chyu and Bergles [6] reported that the HTC initially decreases and then increases as the flow rate increases. On the other hand, Zeng et al. [7] concluded that the effect of flow rate is insignificant at the low saturation temperature, and Parken et al. [8] summarized that the HTCs increase with increasing flow rate. These contradictory descriptions reveal that thermal performance in the falling film evaporation is highly affected by various boundary conditions and thermodynamic states of the working fluid.

Baamrani et al. [13] investigated the effect of heat flux density and flow rate on the evaporation of a liquid film along a vertical plate and reported that the plate surface temperature is higher in cases with lower mass flow rates and higher heat flux density. Zhang et al. [14] investigated the coupling effect of five parameters, including wind velocity, liquid Reynolds number, spray height, heat flux, and spray water temperature, on the evaporation heat transfer of liquid film and proposed empirical heat transfer correlations for circular and half-oval tubes based on their experimental results. Liu et al. [15] conducted a detailed study on the effects of spray density, tube spacing, fluid temperature, and average heat flux on the local heat transfer coefficient outside a horizontal tube. They observed that, starting from the impingement point, the local heat transfer coefficient along the tube circumference first decreases sharply, then gradually declines, and finally shows a slight increase. Zhang et al. [16] examined the falling film heat transfer performance of water on a partially dry-out surface outside a plain copper tube with an outer diameter of 90 mm. The experiments were conducted at a liquid flow density ranging from 0.019 to 0.099 $\text{kg}/\text{m}\cdot\text{s}$ and a heat flux ranging from 0 to 164.1 kW/m^2 . The results showed that the overall heat transfer coefficient is proportional to the fraction of the surface covered by the liquid film.

Studies have been carried out not only on a single tube but also on

tube arrays consisting of multiple tubes [12,17–20]. Zeng et al. [17] and Chyu et al. [18] reported higher HTC in the top row due to liquid impingement on the tube. In contrast, Fujita and Tsutsui [12] observed an increase in HTC from the top tube to the lower tubes, and Bukin et al. [21] did not observe any significant change in HTC with vertical position in the tube row. Moeykens et al. investigated various factors, including feed rates and bundle configurations, that influence the heat transfer performance in spray evaporation systems using R-134a [19] and R-123 [20] refrigerants. Yan et al. [22] investigated the flow behavior, film thickness, and thermal performance of falling water over an array of horizontal tubes and reported that the heat transfer coefficient was the highest at the first tube and the lowest at the bottom tube. In addition to smooth tubes, the heat transfer characteristics of enhanced tubes, such as low-finned [10], Turbo-B [18], and GEWA-T [19], are also investigated. Liu and Yi [10] employed a roll-worked structured surface tube as a low-cost enhanced tube, showing that the tube can greatly enhance evaporation heat transfer for water and R-11 at low and moderate heat fluxes. Zhao et al. [23] experimentally investigated the falling film evaporation of R134a and R123 on the outside of four enhanced horizontal tubes and one smooth tube, reporting that the heat transfer coefficient of R134a is approximately 2–3 times higher than that of R123 under the same test conditions.

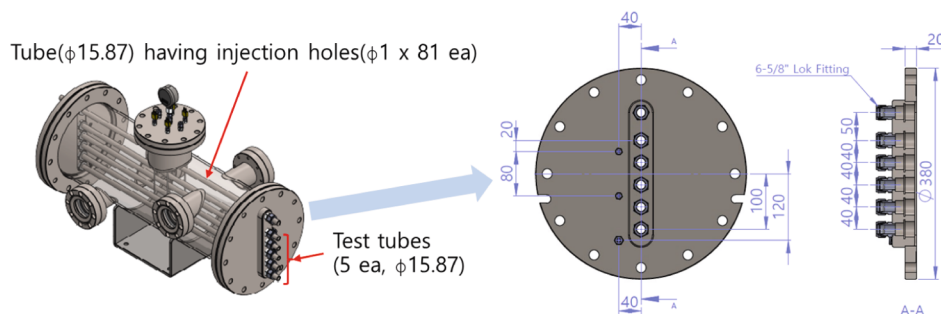
Hsu et al. [24] investigated the heat transfer performance of two pure refrigerants, R32 and R1234yf, as well as near-azeotropic mixtures, R410A, R452B, and R454B, during falling film evaporation on horizontal tubes and reported that R32 exhibited the highest HTC during falling film evaporation on smooth tubes. Chen et al. [25] conducted a three-dimensional numerical simulation to investigate the flow and heat transfer characteristics during falling film evaporation on horizontal tubes, examining the effects of spray densities ranging from 0.0183 to 0.17 $\text{kg}/\text{m}\cdot\text{s}$ on the HTC distribution. Lin et al. [26] attempted to develop a universal correlation for predicting heat transfer coefficients during falling film evaporation on horizontal plain tubes. Zhao et al. [27] provided a comprehensive review of existing correlations related to falling film flow and heat transfer over horizontal tubes and tube

bundles.

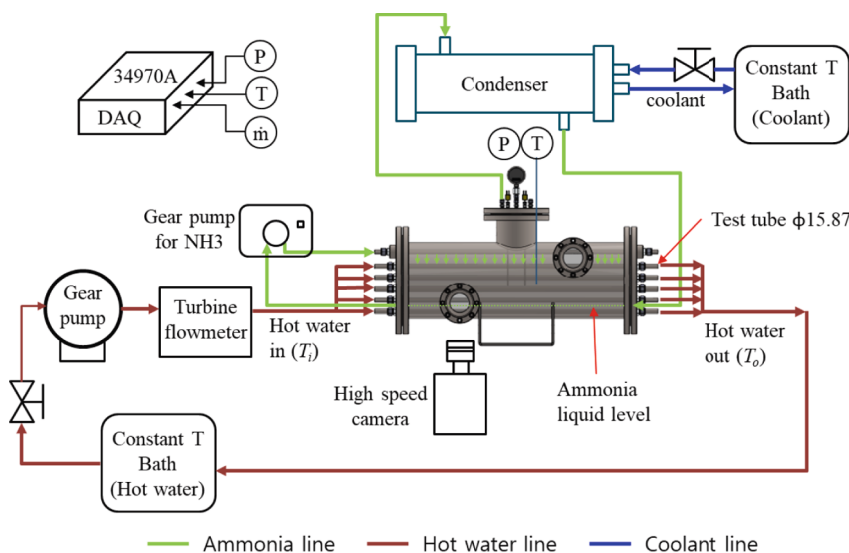
In the case of ammonia, due to its high toxicity and flammability, minimizing the refrigerant charge in heating and cooling systems is a critical consideration, making falling film evaporator technology particularly important. Researchers [24–26] have extensively conducted studies on flooded evaporators using ammonia as a refrigerant, including investigations of both smooth tubes and enhanced tubes. Danilova et al. [28] presented experimental results on the enhanced boiling heat transfer performance of refrigerants R-22 and ammonia using porous metallized coatings in a flooded evaporator. Inoue et al. [29] conducted an experimental study on nucleate pool boiling on a horizontal heated wire in ammonia/water binary mixtures and reported that the heat transfer coefficients of ammonia/water mixtures are significantly smaller than those in single-component refrigerants. Arima et al. [30] measured the boiling heat transfer coefficients of ammonia-water mixtures on a horizontal heating surface and proposed a revised correlation that better fits the data. Meanwhile, rigorous studies on ammonia falling film evaporation were carried out by Lorenz and Yung [31], Chyu et al. [18], and Zeng et al. [17,32,33]. Lorenz and Yung found that liquid film breakdown occurs at the lower rows of tube bundles when the liquid flow rate is too low (below a critical film Reynolds number of 300), resulting in a decrease in HTC. Zeng et al. conducted a series of studies on the heat transfer performance of ammonia spray evaporators by changing various parameters, including saturation temperature, heat flux, spray flow rate, nozzle height, and tube arrangement. Through rigorous studies, they concluded that the

spray evaporation HTC is primarily affected by heat flux and saturation temperature, while spray flow rate has little effect on the evaporation HTC. However, except for the intensive research conducted by Zeng et al., there are not many studies on ammonia falling film evaporation despite the increasing importance of heat pumps using ammonia. Considering the typical operating conditions of adsorption and absorption heat pumps, the heat flux in evaporators generally ranges from a few kW/m^2 to several tens of kW/m^2 , which is relatively low. Under such low heat flux conditions, nucleate boiling is not very active, making evaporation-based heat transfer more dominant.

This study aims to investigate the heat transfer performance of vertically arranged tubes in a falling film evaporator using ammonia refrigerant, in comparison with a flooded evaporator. Through a direct and quantitative comparison with submerged natural convection/pool boiling tubes, the thermal performance differences between the two evaporation modes were more clearly identified. Furthermore, by measuring the heat transfer coefficients of individual tubes at each position and observing the behavior of the ammonia liquid film, the effect of tube position on heat transfer performance was analyzed. Based on the experimental results, an appropriate correlation applicable to various tube positions under low heat flux conditions was proposed. Not only smooth tubes but also enhanced tubes with low fins were examined to analyze the extent of heat transfer performance improvement and the limitations of low-fin enhanced tubes.



(a) Five test tubes for falling film evaporation and one injection tube



(b) Experimental apparatus for falling film evaporation of ammonia

Fig. 1. (a) Tube configuration and (b) experimental apparatus for falling film evaporation of ammonia.

2. Experimental Setup

2.1. Experimental apparatus and test procedure

Fig. 1(a) shows five test tubes for falling film evaporation and one injection tube, all made of stainless steel 316. Five test tubes are vertically aligned with a distance of 40 mm, and the injection tube is located 50 mm above the uppermost of the five tubes. The outer diameter of all tubes is 15.87 mm, and the effective length is 796 mm, which means the length in contact with ammonia. The arithmetic average roughness (R_a) of the smooth tube was 0.296 μm , which was measured using a surface roughness tester (Mitutoyo, SJ-410). The injection tube has 81 injection holes, 1 mm in diameter at the bottom, to supply the jet flow of working fluid downwards to the test tubes. The falling film evaporator consists of these tubes and a large cylindrical chamber that houses them. The chamber has an inner diameter of 254 mm and is equipped with two glass windows, one at the top and one at the bottom, for visual observation. The liquid level of ammonia was controlled to lie between the 4th and 5th test tubes, so that the lowermost test tube (Tube 5) was immersed in the liquid pool of ammonia. Therefore, film evaporation likely occurs from Tube 1 (1st test tube) to Tube 4, while natural convection or pool boiling can occur at Tube 5.

Liquid ammonia is sucked at the bottom of the liquid pool and supplied to the top injection tube with the help of a magnetic gear pump and eventually ejected to the test tubes as shown in Fig. 1(b). The ammonia vapor generated in the falling film evaporator is transported to a shell-and-tube type condenser, where it condenses and returns to the evaporator. The five test tubes are heated with hot distilled water flowing through them, and the hot water flow is controlled using a magnetic gear pump and a turbine flowmeter (Omega, FTB-800). In each of the five test tubes, the flow rate was controlled separately using an individual gear pump and a turbine flowmeter. The inlet temperature of the hot water is adjusted using a constant temperature bath with a temperature stability of ± 0.1 °C. The inlet and outlet temperatures of hot water (T_i and T_o) and the internal saturation temperature (T_{sat}) were measured using an ultra-precise RTD sensor (Omega) with a 1/10 DIN accuracy. The internal pressure of the chamber was obtained using an absolute pressure transducer (Omega, PX409). The data acquisition unit (Keysight, 34970A) acquired all measured values, including temperature, flow rate, and pressure. Photographs of thermal fluid phenomena were captured with a high-speed camera (Phantom, Miro 320S) at a frame rate of 2000 frames per second (fps).

The falling film evaporation experiment for ammonia was conducted at three saturation temperatures: 10, 20, and 30 °C, according to the following procedure. First, the evaporator chamber is evacuated down to a few mTorr using a rotary vacuum pump. Then, liquid ammonia (99.9 %) from the gas cylinder is supplied to the chamber until the lowest tube (Tube 5) is completely immersed in the liquid. The magnetic gear pump for ammonia is turned on, and liquid ammonia circulates and discharges to the test tubes. Initially, the hot water for the evaporator and the cooling water for the condenser are maintained at the same temperature as the predetermined saturation temperature. Then, the hot water temperature is increased by the predefined amount to heat the test tube and cause the film evaporation. Conversely, the cooling water temperature for the condenser is decreased accordingly to maintain the saturation temperature in the chamber. Once the inlet and outlet temperatures of hot water and saturation temperature are stabilized, all measurements, including temperatures, flow rate, and pressure, are stored, and the hot water temperature is increased to the next step. When the hot water temperature reaches the maximum attainable value, the entire experiment is terminated. Two separate experiments were conducted: the first using smooth tubes, and the second using low-fin tubes. In the smooth tube experiment, all five tubes (Tube 1 to Tube 5) were smooth. In the low-fin tube experiment, all five tubes were replaced with low-fin tubes.

2.2. Data reduction and uncertainty analysis

Table 1 shows test conditions for the falling film evaporation of ammonia on smooth and low-fin tubes. The test conditions in Table 1 were set to align with the operating conditions of adsorption heat pumps that utilize low-temperature waste heat—an application of ammonia refrigerant that has recently garnered increasing attention. Adsorption heat pump systems are being developed to supply cooling in the range of 10 to 30 °C to data centers and smart farms by using lower-temperature waste heat as the desorption heat source. Accordingly, the evaporator's saturation temperature was set between 10 and 30 °C. The saturation temperature changes from 10 to 30 °C, corresponding to a change in the saturation pressure of ammonia from 6.15 to 11.7 bar. Table 2 presents the key thermophysical properties of ammonia at saturation temperatures of 10, 20, and 30 °C. Here, ρ denotes the density, c_p the isobaric specific heat, μ the viscosity, k the thermal conductivity, Pr the Prandtl number, and σ the surface tension. The subscripts l and v indicate the liquid and vapor states, respectively.

The ammonia flow rate and heat flux conditions were determined based on the results of previous studies by Fujita and Tsutsui [12], with a focus on the low heat flux range of 0.5–18.2 kW/m², which is suitable for adsorption heat pump applications. The cooling water flow rate was set to ensure a Reynolds number above 3000 (turbulent regime), allowing the application of the Gnielinski correlation. The flow rate of liquid ammonia from the injection tube varies from 0.85 to 2.68 LPM, and the flow rate of hot water is maintained at approximately 3.1 LPM for each test tube. In the case of the smooth tube, the heat input per test tube ranges from 20 W to 721 W, corresponding to a heat flux of 0.5 kW/m² to 18.2 kW/m². In the case of the low-fin tube, the heat flux, based on the base surface area excluding the fin area, ranges from 1.0 kW/m² to 25.8 kW/m². Since this study assumes heat-driven adsorption heat pumps as the primary application, the experiments were conducted under relatively low heat flux conditions, which limits their applicability to systems that require high heat flux. Compared to studies that use electric heaters as the evaporation heat source, this approach has the disadvantages of a narrower heat flux range and somewhat higher uncertainty in the heat transfer coefficient. However, it has the advantage of closely simulating the actual operating conditions of evaporators heated by fluid flowing inside the tubes.

The evaporation heat transfer coefficient (HTC), h_o , for each tube can be obtained from the overall thermal resistance equation in Eq. (1), which comprises three thermal resistances related to internal convection, tube wall conduction, and external evaporation. Fig. 2 shows the schematic drawing of the test tube and the flow structure of the ammonia liquid film. Inlet temperature, outlet temperature, and flow rate (T_i , T_o , and $\dot{m}$) through the test tube are measured in the experiment.

$$\frac{1}{UA} = \frac{1}{h_i A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{1}{h_o A_o} \quad (1)$$

Where U refers to the overall heat transfer coefficient (HTC) and D_o , D_i , k , and L indicate the tube outer diameter, inner diameter, the thermal conductivity of the tube wall, and the length of the test tube, respectively. h_i and h_o refer to the internal convection HTC and the external evaporation HTC, respectively. Overall HTC can be obtained by equating the heat input (Q) calculated from the enthalpy change of hot

Table 1

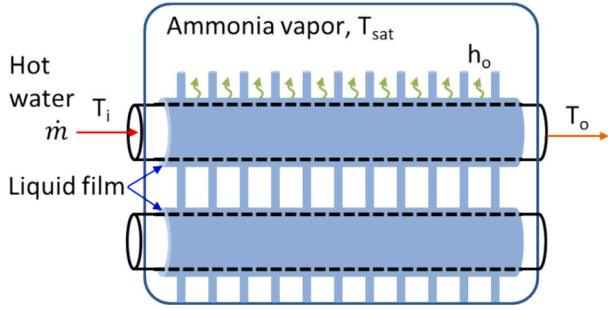
Test conditions for the falling film evaporation of ammonia on smooth and low-fin tubes.

Saturation temperature	10, 20, 30 °C
Saturation pressure	6.15, 8.57, 11.7 bar
Ammonia flow rate	0.85 – 2.68 LPM
Hot water flow rate	3.1 LPM
Heat flux (smooth tube)	0.5 – 18.2 kW/m ²
Heat flux (low-fin tube)	1.0 – 25.8 kW/m ²

Table 2

Thermophysical properties of ammonia at saturation temperatures of 10, 20, and 30 °C.

T_{sat} (°C)	P_{sat} (kPa)	ρ_l (kg/m ³)	ρ_v (kg/m ³)	c_{pl} (kJ/kgK)	μ_l (mPa-s)	k_l (mW/mK)	Pr_l	σ (mN/m)
10	614.8	624.8	4.866	4.668	148.0	502.3	1.376	23.98
20	857.0	610.4	6.698	4.739	133.7	481.0	1.317	21.66
30	1166.5	595.4	9.046	4.826	121.2	458.8	1.275	19.37

**Fig. 2.** Schematic drawing of the test tube and flow structure of the ammonia liquid film.

water through the tube with the overall heat transfer on the tube surface, as shown in Eq. (2).

$$Q = \dot{m}c_p(T_i - T_o) = UA\Delta T_{lm} \quad (2)$$

$$\text{here, } \Delta T_{lm} = \frac{\Delta T_i - \Delta T_o}{\ln(\Delta T_i / \Delta T_o)}$$

$$\Delta T_i = T_i - T_{sat}, \Delta T_o = T_o - T_{sat}$$

Where c_p refers to the isobaric specific heat of hot water and ΔT_{lm} implies the log mean temperature difference through the single test tube. The internal HTC, h_i , is calculated from the Gnielinski correlation [30] for internal convective flow in the transition and turbulent regions, as expressed in Eq. (3).

$$Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (3)$$

$$f = (0.790 \ln Re_D - 1.64)^{-2}$$

$$0.5 \leq Pr \leq 2000, 3000 \leq Re_D \leq 5 \times 10^6$$

The conduction thermal resistance in Eq. (1) can be calculated from the dimensions of the test tube. Now, as the only unknown remaining in Eq. (1) is the last term, the evaporation HTC, h_o , can be obtained.

The inlet temperature of the cooling water to the condenser was controlled to maintain the chamber's saturation temperature at the set value. As a result, the saturation temperature inside the chamber was kept within $\pm 0.2^\circ\text{C}$ throughout the entire experiment. Additionally, all experimental data were averaged over 5 min under steady-state conditions, during which the temperature measurements showed a standard deviation of less than 0.1°C . The measurement uncertainties for the primary dependent variables, heat flux and evaporation HTC, were

Table 3

Accuracy of the measurement sensors and resultant uncertainties.

Independent variables	Sensor	Accuracy
Temperature	Ultra-precise RTD	± 0.03 at 0°C
Pressure	PX409 (OMEGA)	± 0.08 % BSL
Flow rate	Turbine flow meter	± 0.5 % F.S.
Dependent variables	Avg. uncertainties	
Heat flux	4.6 %	
Evaporation HTC	14.6 %	

analyzed according to Moffat's instructions [34]. Table 3 summarizes the accuracy of the present measurement sensors, including temperature sensors (RTD), pressure transducer, and turbine flowmeter. Ultra-precise RTDs have a minimal error as small as $\pm 0.03^\circ\text{C}$ at the reference temperature of 0°C , and the turbine flowmeter has an accuracy of ± 0.5 % in full scale. A rigorous uncertainty analysis yielded average uncertainties of 4.6 % for heat flux and 14.6 % for the evaporation HTC at a 95 % confidence level. Because the uncertainty of HTC is about 14.6 %, which is not particularly large, the analysis of performance differences between tube positions presented in the results, as well as the heat transfer correlation derived from the coefficients, can be regarded as meaningful.

3. Results and discussion

The results are divided into three parts. First, the evaporation heat transfer performance is examined on a smooth tube to investigate how it varies depending on the tube's position. A direct comparison of the evaporation heat transfer coefficients is conducted between the flooded condition (Tube 5) and the falling film condition (Tubes 1–4). Next, the effect of ammonia injection flow rate is analyzed, and an empirical correlation is derived that incorporates the impact of injection flow rate, saturation pressure, and heat flux. Finally, the evaporation heat transfer performance of a low-fin enhanced tube is evaluated and compared to that of the smooth tube.

3.1. Submerged pool boiling versus film evaporation

The upper four test tubes are exposed to film evaporation due to the ammonia liquid film, while the lowest one is submerged in liquid ammonia and affected by flooded evaporation. The term ‘flooded evaporation’ does not seem accurate because the lowest tube is fully immersed, transferring heat by natural convection or pool boiling rather than evaporation. Therefore, in this study, the term ‘flooded evaporation’ was replaced with ‘submerged natural convection/pool boiling’. It is of interest which one shows better heat transfer performance between falling film evaporation at Tube 1 to Tube 4 and pool boiling at Tube 5. Additionally, there can be differences in thermal performance among the four tubes, which are governed by film evaporation.

Fig. 3 shows the evaporation heat transfer coefficient (HTC) on five test tubes for three flow rates ranging from 0.859 LPM (0.00874 kg/s) to 2.68 LPM (0.0272 kg/s) at the saturation pressure of 8.57 bar ($T_{sat} = 20^\circ\text{C}$). Notably, Tube 5 exhibits a much lower HTC than the upper four tubes, indicating that natural convection or pool boiling in the liquid pool is less effective than film evaporation via a thin liquid film. At the low flow rate of 0.859 LPM, Tube 5 shows HTC lower than $1000 \text{ W/m}^2\text{K}$ throughout the heat input range. In contrast, Tube 1 to Tube 4 exhibit higher HTCs ranging from 2422 to $6293 \text{ W/m}^2\text{K}$. At similar heat fluxes of approximately 5000 W/m^2 , the HTC for Tube 5 is $875 \text{ W/m}^2\text{K}$, and the average HTC for the upper four tubes is $3880 \text{ W/m}^2\text{K}$, which is 4.4 times higher than that of Tube 5. Based on this direct comparison, it is certain that heat transfer due to natural convection or pool boiling in Tube 5 is much inferior to that of film evaporation occurring in Tube 1 to Tube 4. Relative comparisons between falling film and submerged natural convection/pool boiling can be found in Chyu and Bergles (1989) and Zeng et al. (2001), all of which, similar to our findings, demonstrate that the heat transfer coefficient of the falling film configuration is

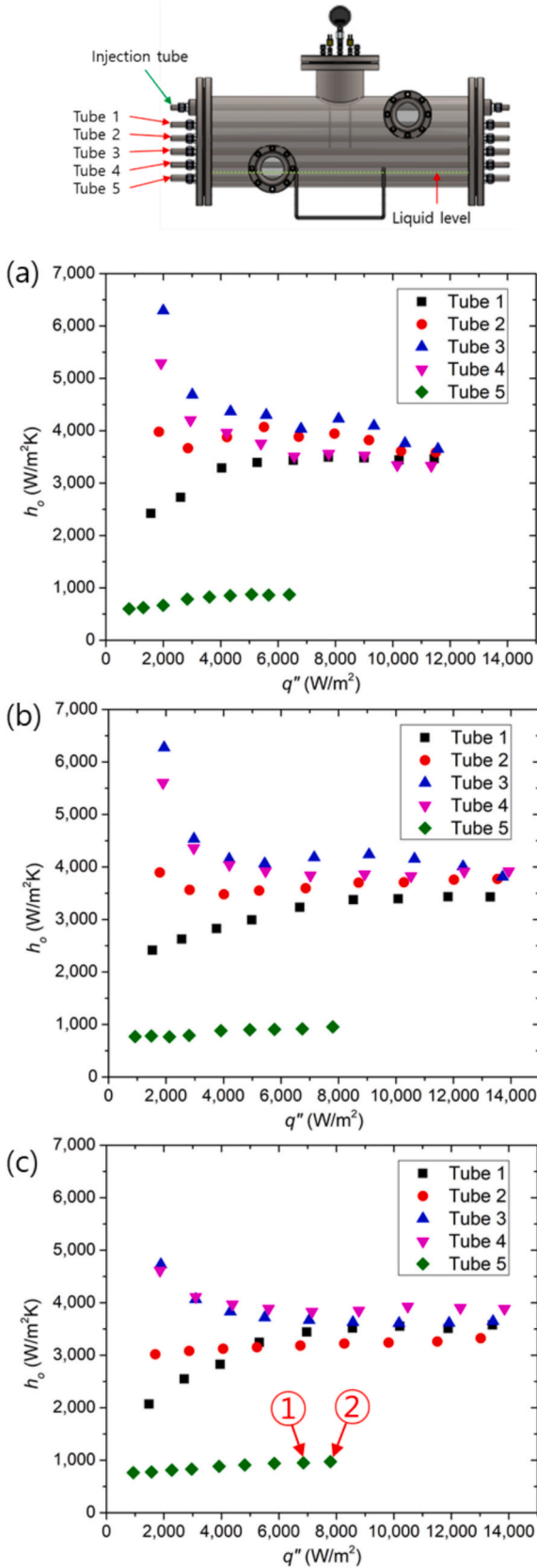


Fig. 3. Evaporation HTC for three flow rates of (a) 0.859 LPM, (b) 1.97 LPM, (c) 2.68 LPM at $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C). HTC is calculated based on D_o .

significantly higher than that of the submerged type.

In Fig. 4, Tube 5, which is submerged in liquid ammonia, transfers heat primarily through natural convection or pool boiling. However, since nucleate boiling is minimal, heat transfer is dominated by natural convection, resulting in a lower heat transfer coefficient for Tube 5. For Tube 4 above it, a thin liquid film can be observed surrounding the tube, and evaporation heat transfer occurs through this thin film, leading to a higher heat transfer coefficient. The thermal performance degradation of Tube 5 is primarily due to the dominant single-phase heat transfer, rather than phase-change heat transfer, in Tube 5 submerged in liquid ammonia. Numerous bubbles should be generated on the tube surface to attain high heat transfer performance in the liquid pool of ammonia. However, bubbles are not seen in most heat flux ranges tested in Tube 5, and single-phase natural convection dominates the submerged tube. It is visible in photographs of fluid flow near Tube 5, taken with a high-speed camera. In Fig. 4(a), only one bubble is generated on the tube surface at the heat flux of 6854 W/m^2 , and no bubbles appear below the heat flux. If no bubbles are generated on the tube surface, the primary heat transfer mechanism of Tube 5 can be considered natural convection. At the next heat flux of 7793 W/m^2 in Fig. 4(b), one more bubble is created on the tube surface. Despite the onset of bubble nucleation, it is natural that the heat transfer performance of Tube 5 is low because most of the tube surfaces are still operating under a natural convection heat transfer mechanism. Kim et al. [32] noted that bubble nucleation is delayed in a highly wetting fluid, such as ammonia, when increasing the heat flux, because most nucleation sites are not activated until the heat flux is sufficiently high.

3.2. Effect of tube location on smooth tubes

The evaporation HTC is also affected by the location of the test tube. As shown in Fig. 3(a-c), the HTC on Tube 1 is lower than on the other three tubes (Tube 2 to Tube 4), especially at the low heat flux region. Surprisingly, the uppermost tube (Tube 1) shows the lowest heat transfer performance despite the direct impingement of the liquid jet. This is not the first report of the lower heat transfer performance of the first tube at film evaporation in the vertical tube array. Fujita and Tsutsui [12] also reported that the first tube among the vertically aligned five tubes exhibits the lowest HTC. They considered that the direct liquid supply from the feeder is responsible for this lower HTC because uniform film flow is not attained on the topmost tube. A closer look at the non-uniform film formation reveals that locally thick liquid films develop on the surface of the first tube, which can be associated with the lower heat transfer performance of Tube 1.

Fig. 5 shows the flow structure of the liquid film discharged from the injection tube to the first test tube for three flow rates ranging from 0.859 to 2.68 LPM. Regardless of the ammonia flow rate, the liquid film is thick at the stagnation point where the liquid jet directly impinges. Film evaporation occurs at the interface between the liquid and vapor phases, while heat conduction is dominant in the liquid film from the tube wall to the liquid-vapor interface. As the liquid film on the stagnation point is thick, there is considerable conduction resistance through the liquid film. However, this stagnation point is just a spot, with minimal effect on the entire HTC. More importantly, the adjacent two jets meet in the middle and form a line of thick liquid film. As shown in the image in Fig. 5, ammonia is injected downward through the bottom holes of the injection tube and impinges on the first tube (Tube 1), forming a liquid film. On the tube surface, a thick liquid film forms at the point where two adjacent ammonia films converge, becoming a factor that reduces heat transfer performance. These thick liquid lines significantly affect the evaporation HTC on Tube 1 because they periodically form around the tube perimeter. Since the liquid film becomes thicker at the point where two adjacent jets meet, this phenomenon is closely related to the design of the liquid distributor. Depending on the spacing of the distributor injection holes, the film formed in the middle can become either thicker or thinner. While a wider spacing between the

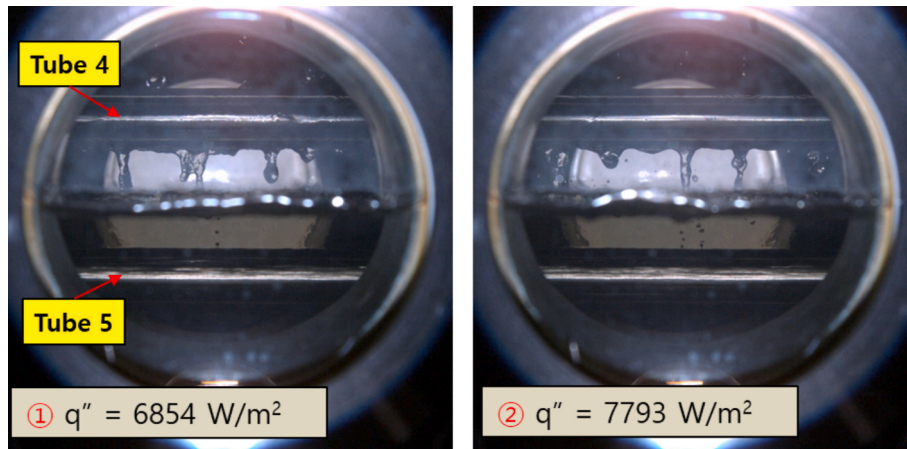


Fig. 4. Bubble nucleation photographs on the submerged tube (Tube 5) at the heat fluxes corresponding to the points ① & ② in Fig. 3(c).

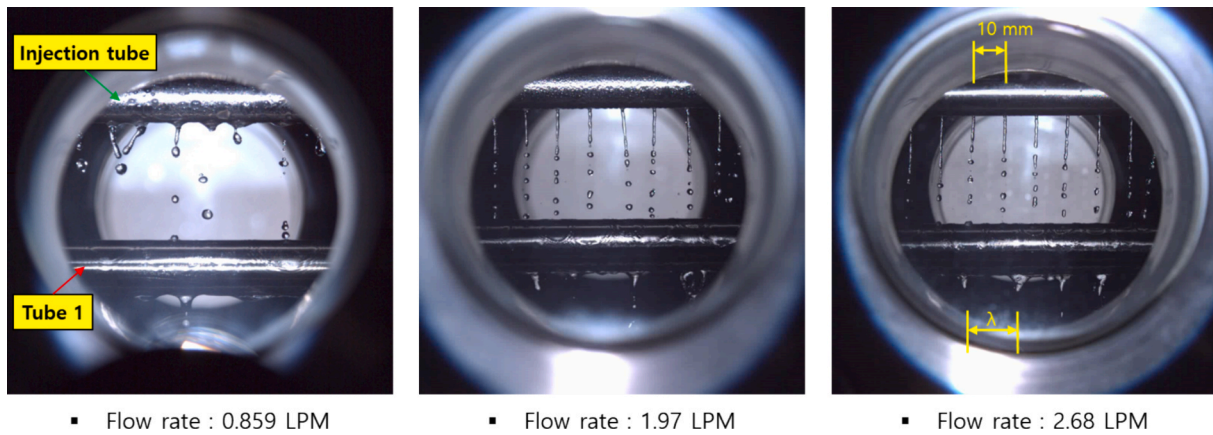


Fig. 5. Flow structure of the liquid film on the Tube 1 according to the flow rate at $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C).

holes may lead to a thinner film at the midpoint, the increased discharge volume from each hole can intensify the non-uniformity of the film at the impingement region. Therefore, it is considered that there exists an optimal spacing of the distributor injection holes that balances heat transfer performance with film uniformity.

Then, it is essential to determine whether the next tube (Tube 2) has the same problem of having a locally thick liquid film on the surface. Unlike the first tube, the downstream liquid column is dominated by the Rayleigh-Taylor instability [35]. The distance between two adjacent columns differs from the distance between the two injection holes (10 mm) and equals the most dangerous wavelength of the liquid-vapor interface, λ_d . The most dangerous wavelength is calculated to be 20.8 mm according to the liquid film instability theory on horizontal tubes (Eq. (4)) [36,37]. In Fig. 5, at a spray flow rate of 2.68 LPM, the measured average spacing between the liquid film columns falling from Tube 1 to Tube 2 is 19.3 mm. This value is found to be like the calculated wavelength of 20.8 mm based on Rayleigh-Taylor instability theory. The spacing between injection holes (s) can be nondimensionalized by the tube diameter (D_o) to define the dimensionless pitch ratio, s/D_o , which is calculated to be 0.63. Similarly, when the most dangerous wavelength, λ_d , derived from Rayleigh-Taylor instability is nondimensionalized by the tube diameter, λ_d is found to be 1.31. It is found that λ_d is about twice as large as s/D_o . In Tube 2, the liquid film columns fall with a spacing nearly twice that of Tube 1, thereby alleviating the formation of locally thick liquid films.

$$\lambda_d = 2\pi \sqrt{\frac{3\sigma}{g(\rho_l - \rho_v)}} \quad (4)$$

More importantly, the drop timings of the liquid columns are not equal, and since two adjacent liquid columns fall on Tube 2 at different times, they are less likely to form thick liquid lines in the middle. A larger distance between the adjacent two liquid columns and inconsistency of their drop timings make the liquid film spread out easily over the entire tube surface, resulting in a thinner and more uniform liquid film. This resulted in higher HTCs on the downward tubes (Tubes 2 to Tube 4) compared to those of the first tube (Tube 1).

Table 4 presents the variation in ammonia flow rate along a vertical tube array at a heat flux of approximately 10,000 W/m² and a saturation pressure of 8.57 bar, as well as the resulting change in the ammonia flow entering each tube. Here, the reduction ratio represents the percentage decrease in flow rate at each tube compared to the initial injection flow rate. Since the entire injection flow enters Tube 1, its reduction ratio is 0 %. As evaporation occurs while the liquid travels downward through the tube array, the ammonia flow rate entering the lower tubes decreases. Notably, when the ammonia injection flow rate is at its lowest—0.00874 kg/s (0.859 LPM)—the flow rate entering Tube 4 is reduced by nearly 12 % compared to that entering Tube 1. This decrease in ammonia flow implies a similar reduction in the average thickness of the liquid film surrounding Tube 4. As long as dryout

does not occur, the thinner film leads to lower thermal resistance, resulting in a higher evaporation heat transfer coefficient for Tube 4 than for Tube 1. However, as the heat input increases, the liquid film may evaporate to the point that it can no longer completely wet the tube surface, leading to localized dryout. This dryout is more likely to occur first at Tube 4, where the film is already thinner. In the above discussion, the differences in heat transfer performance with tube position were

Table 4

Evaporation rate through each tube in a vertical tube array and the corresponding flow rate reduction at the heat flux around 10,000 W/m² and $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C).

Injection flow rate (kg/s)	Tube #	Q (W)	Tube inlet flow rate (kg/s)	evaporation rate (kg/s)	reduction ratio (%)
0.0087	Tube 1	405.7	0.00874	0.00034	0.0 %
	Tube 2	408.3	0.00840	0.00034	3.9 %
	Tube 3	413.7	0.00805	0.00035	7.8 %
	Tube 4	403.0	0.00771	0.00034	11.8 %
0.0201	Tube 1	399.9	0.02008	0.00034	0.0 %
	Tube 2	407.9	0.01975	0.00034	1.7 %
	Tube 3	422.3	0.01940	0.00036	3.4 %
	Tube 4	417.6	0.01905	0.00035	5.2 %
0.0272	Tube 1	405.1	0.02724	0.00034	0.0 %
	Tube 2	389.7	0.02690	0.00033	1.3 %
	Tube 3	404.0	0.02657	0.00034	2.5 %
	Tube 4	416.1	0.02623	0.00035	3.7 %

explained in terms of the behavior of the ammonia liquid film on the tube surface. However, to support this qualitative analysis, quantitative measurements of the liquid film thickness are required. Furthermore, considering that the liquid film behavior is strongly influenced by the wettability characteristics of ammonia, additional studies on ammonia wettability on tube surfaces should be conducted in future work.

3.3. Effect of flow rate on the evaporation HTC

To begin with, the effect of flow rate on the evaporation HTC was examined before deriving the empirical correlation for the evaporation heat transfer of ammonia. The ammonia flow rate discharged from the injection tube was changed from 0.852 to 2.68 LPM at three saturation pressures of 6.15, 8.57, and 11.7 bar ($T_{sat} = 10, 20$, and 30 °C). Fig. 6 shows the evaporation HTCs on the five test tubes from Tube 1 to Tube 5 at the saturation pressure of 11.7 bar ($T_{sat} = 30$ °C). It seems that the effect of flow rate is minimal at this relatively high saturation pressure and temperature. Only Tube 4 exhibits slightly lower HTCs at the low flow rate of 0.852 LPM. Tube 4 is the last one among the four tubes that are not submerged. The discharged liquid jet passes through three upper tubes to reach Tube 4, where it is transformed into a liquid column by the cylindrical tubes. As already analyzed in the previous section, there are temporal differences in the dropping of the liquid column. This time difference can cause some areas of the tube surface to dry without being covered by a layer of liquid film, which can degrade heat transfer performance. Although local dry-out was not clearly observed in our footage, it is considered the most plausible cause for the reduced heat transfer performance of Tube 4 under low flow rate conditions. Despite a slight difference in Tube 4, the effect of flow rate on the HTCs at $T_{sat} = 30$ °C is not that large, and the difference lies within ± 8 % of the average value.

In comparison, the effect of flow rate increases as the saturation temperature (pressure) decreases, as shown in the HTC graphs at $T_{sat} = 10$ °C of Fig. 7. As the saturation temperature decreases from 30 to 10 °C, surface tension of liquid ammonia increases from 19.4 to 24.0 mN/m and dynamic viscosity also increases from 121 to 148 μ Pa·s. Both properties increase by approximately 22 to 24 % compared to those at

$T_{sat} = 30$ °C. According to the conventional Nusselt theory, the thickness of the liquid film flowing around a cylinder is proportional to one-third power of the viscosity; therefore, as the saturation temperature decreases, the film thickness increases. Based on the Rayleigh–Taylor instability theory analyzed earlier, an increase in surface tension leads to an increase in the most dangerous wavelength of the liquid–vapor interface. In addition, the wetting capability on the tube surface deteriorates at low saturation temperatures due to an increase in surface tension and viscosity. The poor wettability hinders the tube surface from being wetted by liquid ammonia, resulting in poorer heat transfer performance, especially at low flow rates. The reduction in the HTCs for Tube 4 at the flow rate of 0.925 LPM is as much as 20 % compared to the average value. While, at higher flow rates the tube surface gradually becomes fully wet. Accordingly, at lower saturation temperatures, the difference in heat transfer performance depending on the spray flow rate becomes more significant. It is also noteworthy that the effect of flow rate on HTC is minimal at Tube 5, since Tube 5 is submerged in liquid ammonia and controlled by natural convection, which is not significantly affected by its properties.

3.4. Deriving empirical correlations on smooth tubes

As a next step, empirical correlations for evaporation heat transfer of ammonia were derived based on measured data in the present study. Previously, Zeng et al. [17] proposed a heat transfer correlation including the effects of flow rate, saturation pressure, and heat flux for the sprayed ammonia evaporation in the following form.

$$Nu = 0.0495Re_f^{-0.00399}Pr^{0.209}P_{red}^{0.261}\Theta^{0.722} \quad (5)$$

$$\text{here, } Re_f = 4\Gamma/\mu, P_{red} = P_{sat}/P_{crit}, \Theta = q''D/(T_{crit} - T_{sat})k$$

Where Re_f , P_{red} and Θ refer to the film Reynolds number, reduced pressure of saturated ammonia, and dimensionless heat flux on the tube surface, respectively. Γ represents the liquid mass flow rate per unit length of tube (each side). Fujita and Tsutsui [12] found that the first tube in the vertical tube array showed a slight dip in the HTC compared to the other tubes below. They suggested two correlations: one for the first tube and one for the lower tubes.

$$Nu = (Re_f^{-2/3} + 0.008Re_f^{0.3}Pr^{0.25})^{1/2} \text{ for the 1st tube (6).}$$

$$Nu = (Re_f^{-2/3} + 0.01Re_f^{0.3}Pr^{0.25})^{1/2} \text{ for the lower tubes (7).}$$

As Tube 1 in the present experiment also showed slightly lower HTC than other tubes (Tube 2–4), it is appropriate to derive different correlations for Tube 1 and Tube 2–4. In addition, the empirical correlation should include both reduced pressure and dimensionless heat flux terms, like Zeng et al.'s correlation. Therefore, the empirical correlation for falling film evaporation on a vertical tube array is suggested to be as follows.

$$Nu = CRe_f^{n_1}Pr^{0.209}P_{red}^{n_2}\Theta^{n_3} \quad (8)$$

Where C is a constant and n_1 , n_2 , and n_3 are the exponents for the Reynolds number, reduced pressure, and dimensionless heat flux, respectively. The film Reynolds number was included to account for the effect of the injected ammonia flow rate on heat transfer, while the Prandtl number was included to reflect the relationship between heat and mass transfer in the evaporation process. The reduced pressure represents the influence of saturation temperature under conditions where the saturation temperature varies, as in the present experiments. The dimensionless heat flux was incorporated to capture the variation in heat transfer behavior depending on the applied heat flux. The exponent for the Prandtl number was preserved as 0.209, the same as Zeng et al.'s correlation, because only ammonia was employed, and the variation of the Prandtl number was minimal in the present study.

Fig. 8(a) depicts all the measured data for Tube 1, Tube 2–4, and Tube 5, along with trend lines obtained from the two previous

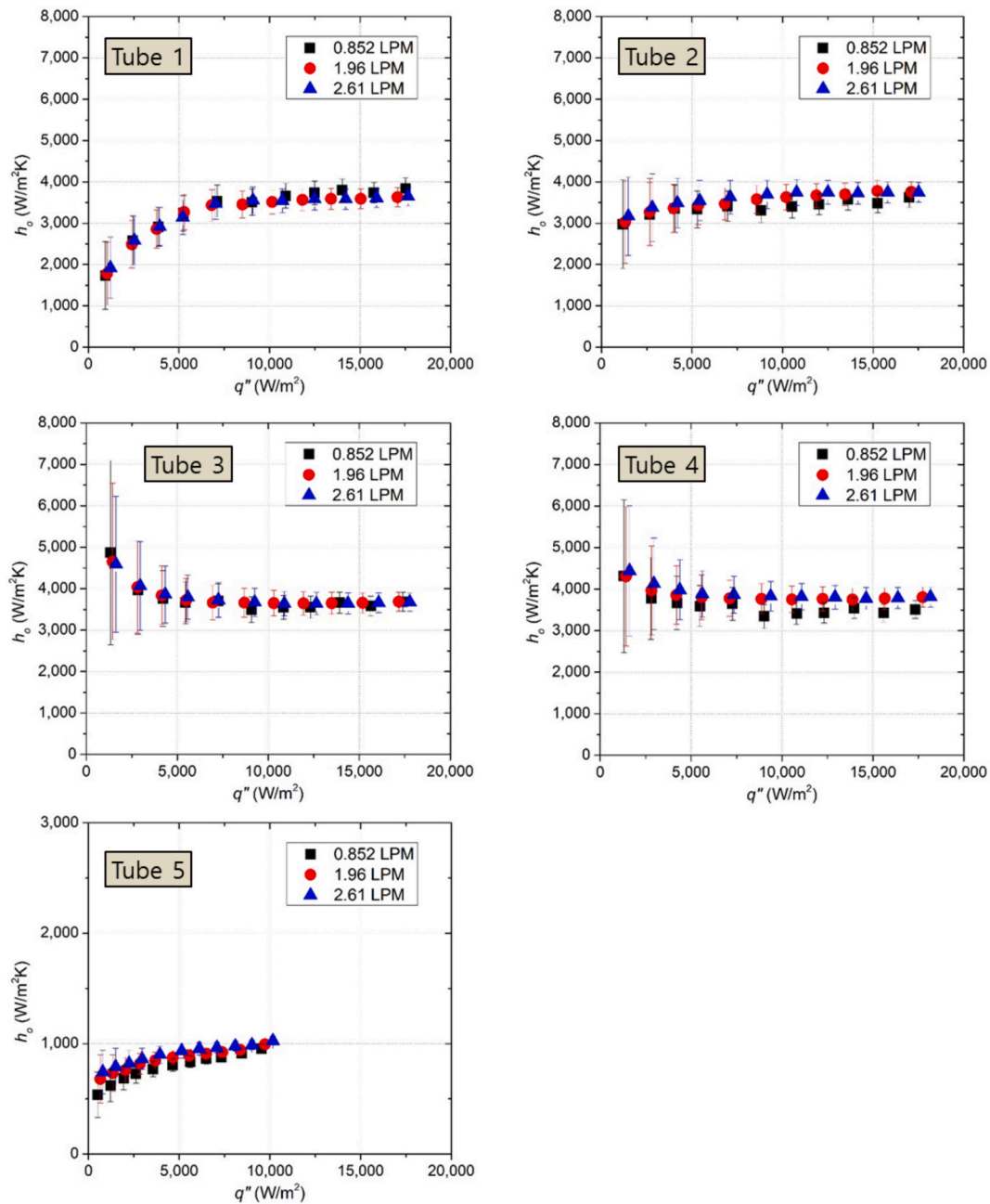


Fig. 6. Evaporation HTC on the five test tubes from Tube 1 to Tube 5 at $P_{sat} = 11.7$ bar ($T_{sat} = 30$ °C). HTC is calculated based on D_o .

correlations in the Nu versus θ graph. Notably, the two correlations deviate considerably from the measured data, and all the measured data for the falling film evaporation lie between them. Fujita and Tsutsui's correlation overestimates the HTC, while Zeng et al.'s one underestimates it. When comparing the measured Nusselt numbers of Tubes 1 to 4 in the film evaporation region with the two correlations (Fujita and Tsutsui, Zeng et al.), the MAPE (Mean Absolute Percentage Error) is as high as 125 % and 62 %, respectively. In the study by Fujita and Tsutsui, the correlation was derived from experimental data obtained with a single working fluid, R11, making it difficult to effectively account for the influence of the Prandtl number. Since the correlation was based on R11 data, which generally exhibits relatively high Nusselt numbers, the predicted heat transfer performance for ammonia tends to be overestimated. In contrast, the study by Zeng et al. was conducted in a low saturation temperature range of -23.3 °C to 10 °C, and the correlation was therefore derived from data corresponding to relatively low

heat transfer coefficients. Furthermore, because the exponent of the dimensionless heat flux term in Eq. (5) is large, the correlation is highly sensitive to heat flux; thus, as in the present study, it can underpredict heat transfer performance in the low heat flux region. As shown in Fig. 8 (a), the predictions from the Zeng et al.'s correlation increase rapidly with heat flux, and when the dimensionless heat flux (θ) reaches about 6, the predicted values approach the measurements obtained in this study.

The HTC (Nusselt number) for Tube 1 increases with the heat flux, and the HTC for Tube 2–4 shows a decreasing trend with the heat flux. However, when the dimensionless heat flux is higher than 2.0, the difference between Tube 1 and Tube 2–4 becomes minimal, and the effect of heat flux on the HTC is diminished. Accordingly, two separate correlations for Tube 1 and Tube 2–4 were devised at the low heat flux region ($\theta \leq 2.0$), and a unified correlation excluding the θ dependency for Tube 1–4 was derived at the high heat flux region ($\theta > 2.0$). The

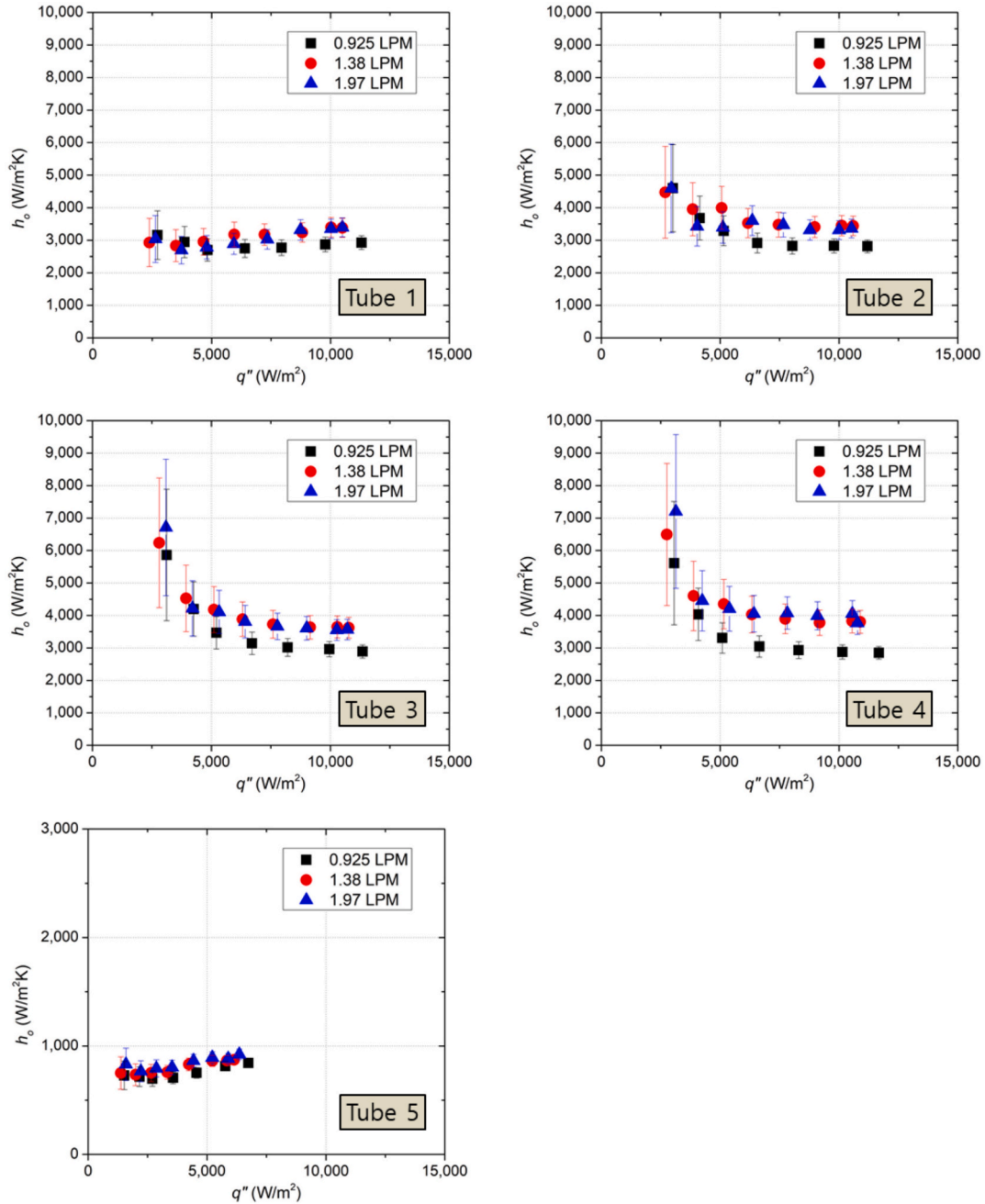


Fig. 7. Evaporation HTCs on the five test tubes from Tube 1 to Tube 5 at $P_{sat} = 6.57$ bar ($T_{sat} = 10$ °C). HTC is calculated based on D_o .

three correlations are presented in Equations (9–11). It can be observed that all three correlations commonly exhibit a small exponent for the Reynolds number. The relatively small exponent value of the Reynolds number in the derived correlation indicates that, within the tested range, the effect of ammonia injection flow rate on heat transfer is not significant.

$$Nu = 0.0905 Re_f^{-0.0376} Pr^{0.209} P_{red}^{-0.0906} \Theta^{0.259} \text{ (for Tube 1, } \Theta \leq 2.0) \quad (9)$$

here, $C = 0.0905, n_1 = -0.0376, n_2 = -0.0906, n_3 = 0.259$

$$159 \leq Re_f \leq 565, 0.053 \leq P_{red} \leq 0.103$$

$$Nu = 0.0842 Re_f^{-0.00253} Pr^{0.209} P_{red}^{-0.188} \Theta^{-0.185} \text{ (for Tube 2–4, } \Theta \leq 2.0) \quad (10)$$

here, $C = 0.0842, n_1 = -0.00253, n_2 = -0.188, n_3 = -0.185$

$$157 \leq Re_f \leq 565, 0.053 \leq P_{red} \leq 0.103$$

$$Nu = 0.114 Re_f^{0.0520} Pr^{0.209} P_{red}^{0.0987} \text{ (for Tube 1–4, } \Theta > 2.0) \quad (11)$$

here, $C = 0.114, n_1 = 0.0520, n_2 = 0.0987$

$157 \leq Re_f \leq 550, 0.054 \leq P_{red} \leq 0.103, 2.0 \leq \Theta \leq 6.1$ Fig. 8(b) shows comparison between all measured Nu for Tube 1–4 (film evaporation) and the predicted Nu using Eq. (9–11). Compared with the previous correlations, which deviate substantially from the measurements, the Nusselt numbers predicted by Eqs. (9–11) agree relatively well with the experimental data. In the correlation for Tubes 2–4 (Eq. (10)), some points exhibit larger deviations, which is because Eq. (10) cannot fully capture the high Nusselt numbers that occur at small Θ values in Fig. 8 (a). MAPEs (Mean absolute percentage error) for Eq. (9–11) from the measured data were calculated as 5.8 %, 11.1 %, and 5.0 %, respectively. Because the heat flux on the tube surface is not known until the

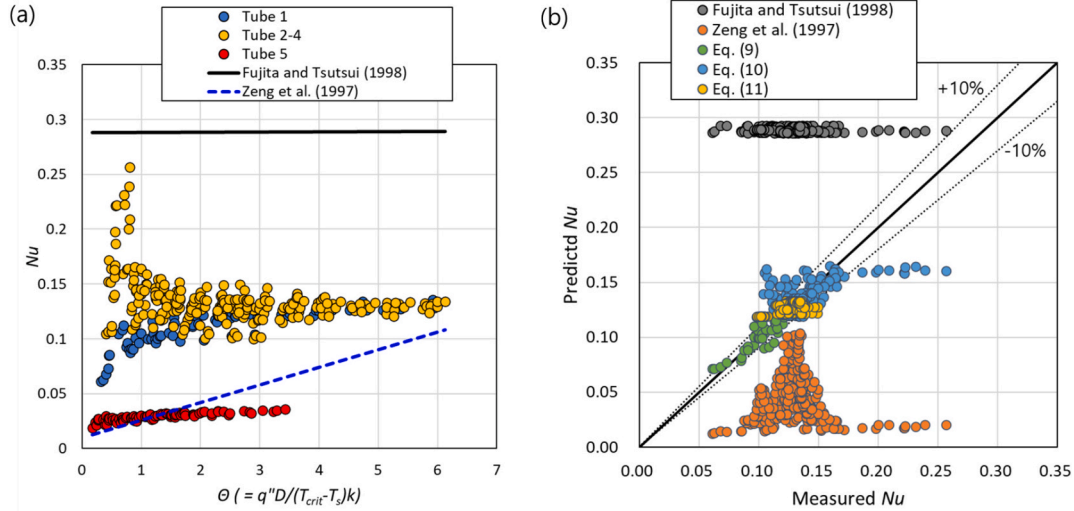


Fig. 8. (a) All the measured data for Tube 1, Tube 2–4, and Tube 5 along with two previous correlations in the Nu versus θ graph, and (b) Measured Nu versus Predicted Nu using Eq. (9–11).

wall HTC is obtained in designing the falling film evaporator, Eq. (11) can give a reasonable first estimation of the evaporation HTC regardless of the tube location. The proposed correlations are applicable to the tested range of 10–30 °C and low heat flux (0.5–18.2 kW/m²), and further validation is required in the higher heat flux region where nucleate boiling occurs.

In this study, a spray tube with 1 mm-diameter holes arranged at 10 mm intervals was used. Since the distributor geometry is closely related to the uniformity of the liquid film thickness, further studies on the geometric parameters of the spray tube are required. In addition, because the Weber number and Bond number were not considered in deriving the heat transfer correlation based on the experimental data, the prediction error of the correlation may increase when applied outside the tested experimental conditions.

3.5. Falling film evaporation on low-fin tubes

Enhanced tubes, such as low-fin tubes, are typically used to improve the thermal performance of the evaporator by incorporating fin structures into the smooth tube surface. In the present study, the low-fin tube, one of the most common enhanced tubes, was selected, and its thermal performance in falling film evaporation was examined in comparison with that of the smooth tube. Table 5 shows the important dimensions of the low-fin tube compared with those of the smooth tube. As the fin structure is formed by the rolling process of the smooth tube, the fin diameter (D_f), including the fin length, is approximately equal to the outer diameter (D_o) of the smooth tube, and the outer diameter of the low fin tube is reduced to 13.07 mm. The evaporation HTC on the low-fin tube was obtained using the base area calculated from this outer diameter, not the fin diameter including the fins, because thermal conduction resistance through the tube wall in Eq. (1) can be adequately estimated using the reduced outer diameter. The fin height for the low-fin tube is 1.33 mm, and the number of fins is 26 per inch. The total surface area of the low-fin tube per unit length was calculated to be 0.1477 m²/m, which is 2.96 times larger than that of the smooth tube.

Table 5
Important dimensions of the low-fin tube compared with those of the smooth tube.

	Material	D_i	D_o	t_w	D_f	h_f	Number of fins (#/inch)	Surface area per unit length (m ² /m)
Smooth	Stainless steel 316	13.39	15.87	1.24	—	—	—	0.04986
Low-fin	Stainless steel 439	11.81	13.07	0.63	15.73	1.33	26	0.1477 (2.96 × $A_{t,smooth}$)

Fig. 9 shows the evaporation HTC on the low-fin tube array at three ammonia flow rates from 0.859 to 2.58 LPM at the saturation temperature of 20 °C ($P_{sat} = 8.57$ bar). The HTC on the low-fin tubes are higher than those on the smooth tubes, but the overall trend is similar to that of the smooth tubes. The uppermost tube (Tube 1) exhibits the lowest HTC because jet flow directly impinges on the first tube, forming a locally thick liquid film layer on the tube surface, similar to the smooth tube. The flow behavior of the ammonia liquid on the first tube (Tube 1) is illustrated in Supplement S1, which features photographs of the liquid jet and film flow captured by a high-speed camera. As described in Section 3.2, the heat transfer performance improves as the flow proceeds downward to the lower tubes (Tubes 2–4), where the liquid film becomes thinner and more uniform. In contrast, Tube 5, which is fully submerged, exhibits a significantly lower HTC. From the comparison of HTCs at three flow rates in Fig. 9, it is also noteworthy that the effect of the flow rate on the HTC of the low-fin tubes is minimal.

From Fig. 9, the HTC obtained by employing the outer diameter, D_o , for the low-fin tube is higher than that for the smooth tube. However, since D_o of the low-fin tube is smaller than that of the smooth tube, it is more appropriate to compare not the HTC (h_o) itself, but the thermal conductance, $h_o A_o$, for the thermal performance comparison. The thermal conductance is related to the calculation of the overall heat transfer coefficient (U) as shown in Eq. (1). Fig. 10 shows the evaporation thermal conductance of the low-fin tube along with that of the smooth tube at the saturation temperature of 20 °C and flow rate of 1.91 LPM. All low-fin tubes exhibit better thermal performance than smooth tubes, regardless of the location from Tube 1 to Tube 5. As described in Section 3.2, Tubes 1 and 2 in the upper region show an increase in HTC as the heat flux increases due to the thinning of the liquid film. In contrast, Tubes 3 and 4 in the lower region exhibit a decrease in HTC with increasing heat flux, which is attributed to the formation of local dryout regions.

The enhancement factor is defined as the ratio of the thermal conductance of the low-fin tube to that of the smooth tube.

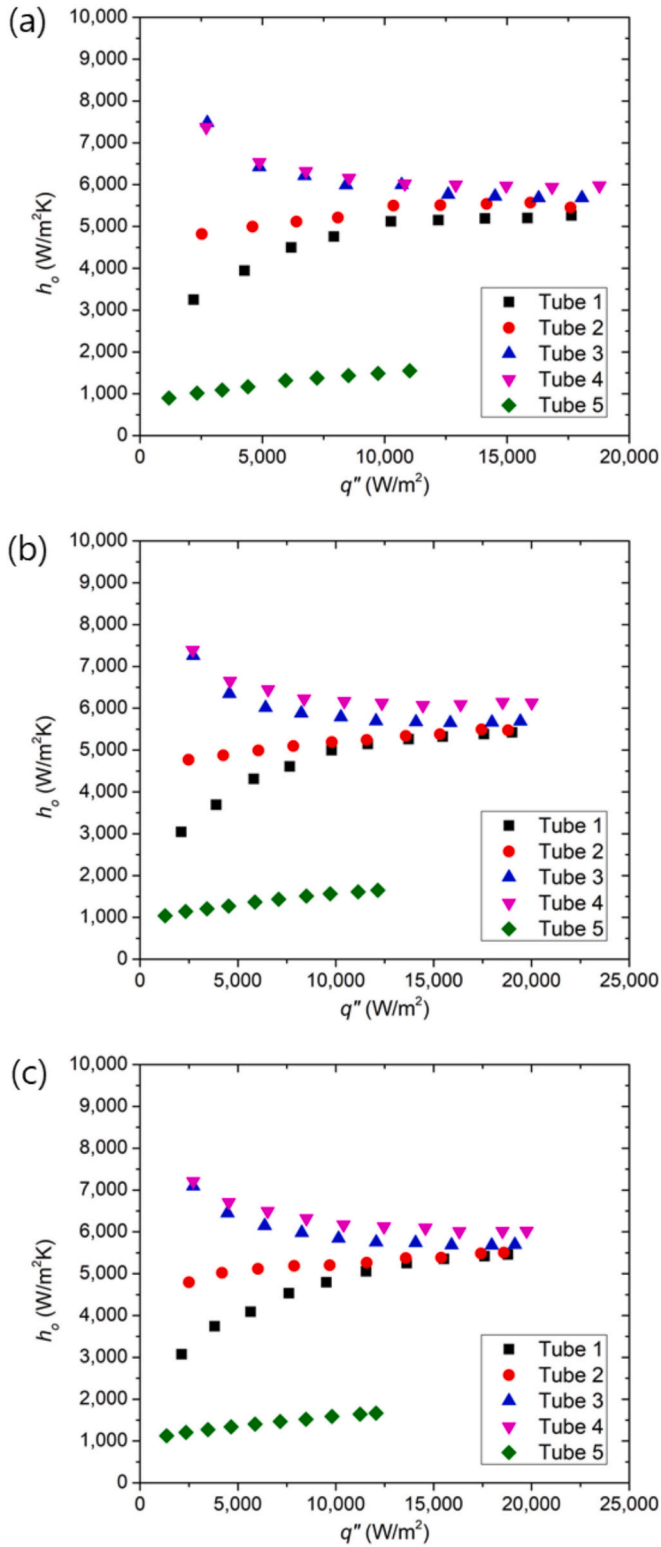


Fig. 9. Evaporation HTC on the low-fin tube array at three ammonia flow rates from 0.859 to 2.58 LPM at $P_{sat} = 8.57$ bar ($T_{sat} = 20$ °C). HTC is calculated based on D_o .

$$\text{Enhancement factor} = \frac{(h_o A_o)_{\text{low-fin}}}{(h_o A_o)_{\text{smooth}}} \quad (12)$$

The enhancement factor increases for the falling film evaporation region (Tube 1–4) as the liquid film flows downward from 1.14 for Tube 1 to

1.34 for Tube 4. It is thought that a thinner and more uniform liquid film at Tube 4 helps utilize the low-fin tube's larger surface area for heat transfer. Similarly, the low-fin tube exhibits better thermal conductance with an enhancement factor of 1.20 than smooth tubes in the natural convection or pool boiling region (Tube 5). This is because the surface area enhancement of the low-fin tube is effective for natural convection or pool boiling heat transfer.

Table 6 summarizes the enhancement factors for all cases tested at three saturation temperatures (pressures) and various flow rates. Overall, the enhancement factor increases with the saturation temperature for the falling film evaporation region. The averaged enhancement factor for Tube 1 to Tube 4 at the saturation temperature of 10 °C is merely 1.066, while that at the saturation temperature of 30 °C is as high as 1.325. Although the area enhancement due to the low fins is as large as 2.96 times that of the smooth tube, the annular fins are responsible for blocking the spreading of the ammonia liquid film over the entire tube surface. It is important that the liquid ammonia spreads well over the entire tube surface; however, in the low-fin tube, the fins are oriented perpendicular to the spreading direction of the liquid film, which hinders its spreading.

The low-fin tube has two competing factors: pros increasing the heat transfer area and cons preventing the liquid from spreading out in the axial direction. The latter factor (obstacles to spreading) is likely strengthened at the low saturation temperature because the viscosity and surface tension increase with decreasing temperature. As observed in the smooth tube, higher viscosity thickens the liquid film, which slows spreading, while higher surface tension worsens wettability on the tube surface, thereby weakening the spreading of the liquid film. The poor thermal performance of the low-fin tube at the saturation temperature of 10 °C is due to the greater hindrance to film spreading caused by these effects. Although a 20–30 % enhancement of thermal performance by applying the low-fin tube is not insignificant, it is expected that further improvement will be achieved by employing fin structures designed more elaborately with an axial flow passage for liquid spreading. The limited heat transfer enhancement observed with the enhanced tube is attributed to its geometry not being optimized for falling film conditions. It is suggested that further studies are needed to evaluate various types of enhanced tubes to maximize the evaporation heat transfer performance under falling film conditions.

4. Conclusions

Falling film evaporation heat transfer of ammonia was investigated on vertically aligned five tubes at the saturation temperature from 10 °C to 30 °C. The following conclusions were drawn from the measured data and visual observation.

- From the direct comparison between the upper four tubes affected by film evaporation and the lowermost tube submerged in the liquid ammonia, it is seen that the HTC of falling film evaporation is much higher than that of natural convection or pool boiling at the low heat flux region tested (4.4 times higher at $q'' = 5$ kW/m²).
- The uppermost tube (Tube 1) shows the lowest heat transfer performance among the four tubes because a locally thick liquid film is formed on both the stagnation point of a jet and the middle line between two adjacent jets.
- Three empirical correlations for the falling film evaporation on a vertical tube array are suggested according to the tube location and dimensionless heat flux, which show MAPES of 5.8 %, 11.1 %, and 5.0 %, respectively. The proposed correlations are applicable to the tested range of 10–30 °C and low heat flux (0.5–18.2 kW/m²).
- Thermal performance is improved by 10–30 % on the low-fin tube compared with the smooth tube, and the enhancement is more pronounced at the lower tube (Tube 4) and high saturation temperature.

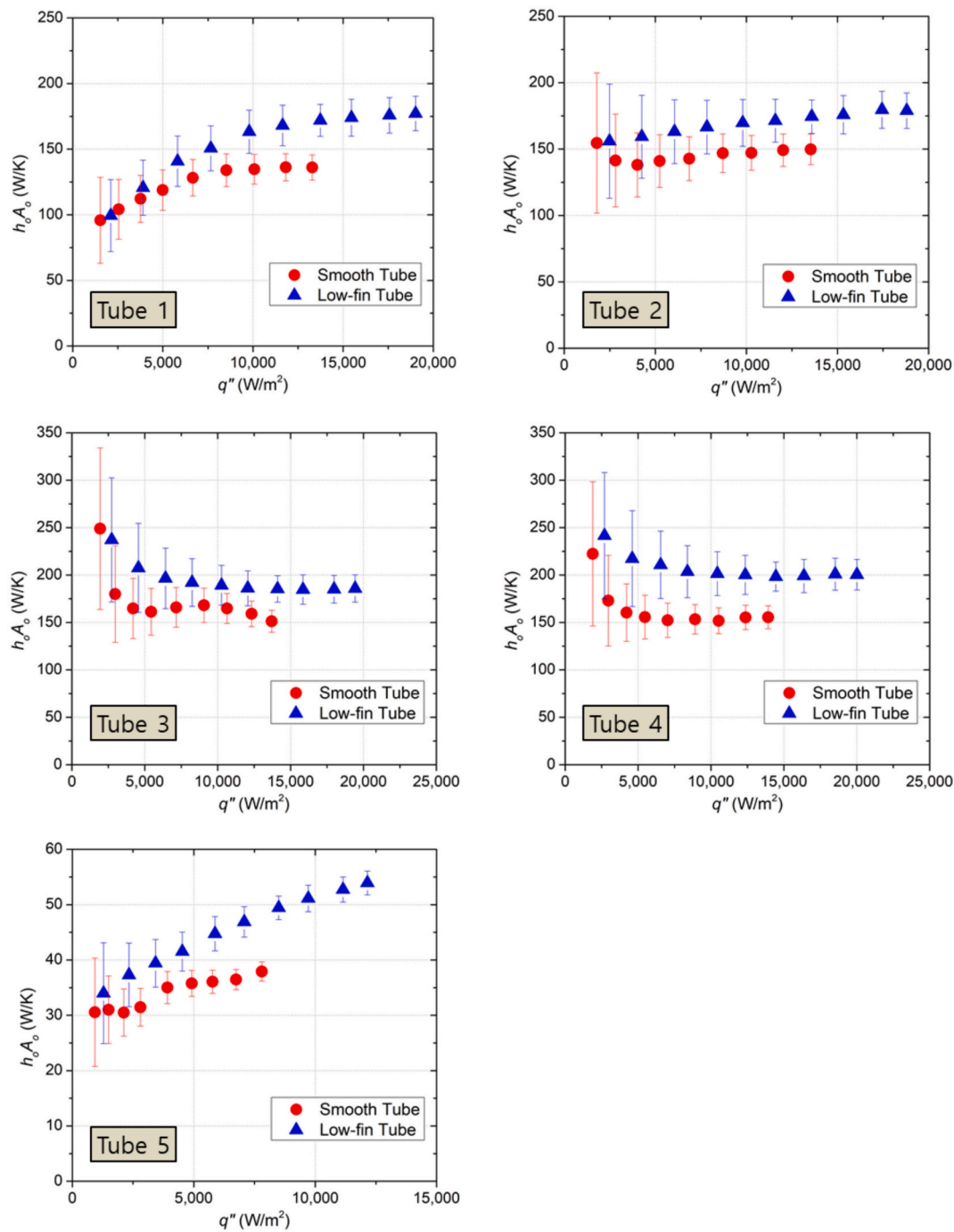


Fig. 10. Evaporation thermal conductance of the low-fin tube along with that of the smooth tube at $T_{sat} = 20\text{ }^{\circ}\text{C}$ and flow rate of 1.91 LPM.

Table 6
Enhancement factors for all cases tested at three saturation temperatures and various flow rates.

$T_{sat,nominal}$ ($^{\circ}\text{C}$)	10			20			30		
P_{sat} (bar)	6.09	6.14	6.15	8.54	8.55	8.55	11.6	11.6	11.7
Flow rate (LPM)	0.925	1.36	1.99	0.863	1.91	2.58	0.925	1.91	2.61
Tube #									
1	1.05	0.91	0.94	1.1	1.14	1.1	1.15	1.29	1.18
2	1.14	0.91	0.98	1.12	1.14	1.32	1.31	1.32	1.27
3	1.31	1.06	1.09	1.25	1.2	1.36	1.33	1.4	1.37
4	1.31	1.05	1.04	1.44	1.34	1.35	1.42	1.46	1.4
5	1.21	1.15	1.17	1.17	1.2	1.23	1.29	1.3	1.29

This study examines the evaporation heat transfer of ammonia under low heat flux conditions and provides correlations applicable to the

design of evaporators for adsorption or absorption heat pumps. By proposing appropriate heat transfer correlations according to heat flux

ranges and tube positions, the results of this study can be practically applied to evaporator design. In the higher heat flux region where nucleate boiling occurs, the validity of these correlations requires further verification. In addition, although the behavior of the ammonia liquid film was qualitatively observed and analyzed through visualization, future studies need to be accompanied by quantitative analyses such as measuring film thickness.

CRedit authorship contribution statement

Jin Sub Kim: Writing – original draft, Methodology, Conceptualization. **Wookyoung Kim:** Writing – review & editing, Investigation. **Hak Soo Kim:** Validation, Formal analysis. **Young Kim:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tsep.2026.104483>.

Data availability

Data will be made available on request.

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